ECON2050 Tutorial 11

Optimization IV: Nonlinear Programming, Endgame

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Xavier Xi Zhang

School of Economics, The University of Queensland

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Today's Plan

- Tutorial questions: Q1–Q3
- Assignment help.

Tutorial Q1

Question 1

Consider the nonlinear programming problem

$$\max_{x,y\in\mathbb{R}} \quad 2x - 2y^2$$

subject to

$$2x - y \leq 5, \qquad y \leq 4, \qquad x \geq 0, \quad y \geq 0.$$

(a) Try to solve the problem (at least roughly) graphically.

(b) Solve the problem using the KKT approach.

Recap: KKT via the reduced Lagrangian

Reduced Lagrangian

Without explicit multipliers on the nonnegativity constraints, set

$$\widetilde{\mathcal{L}}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) - \sum_{j=1}^{m} \lambda_j g_j(\boldsymbol{x}).$$

Equivalent KKT conditions

$$\frac{\partial \widetilde{\mathcal{L}}}{\partial x_i} \leq 0, \qquad x_i \frac{\partial \widetilde{\mathcal{L}}}{\partial x_i} = 0, \qquad x_i \geq 0;$$
$$\frac{\partial \widetilde{\mathcal{L}}}{\partial \lambda_j} \geq 0, \qquad \lambda_j \frac{\partial \widetilde{\mathcal{L}}}{\partial \lambda_j} = 0, \qquad \lambda_j \geq 0.$$

All expressions are evaluated at a candidate $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$.

Recap: When are KKT conditions necessary?

Sufficient hypotheses for necessity

If $\boldsymbol{x}^*$ is optimal, KKT conditions are necessary when either:

1. every constraint function g_j is **affine**; or
2. the gradients of the constraints **binding at $\boldsymbol{x}^*$**, including binding nonnegativity constraints, are linearly independent.

Implication

Under these hypotheses, every optimal solution must be among the KKT candidates. Without them, KKT conditions need not be necessary.

Recap: Finding solutions using KKT

When existence is already established

1. Check whether the constraint functions are affine. If not, check linear independence of gradients of binding constraints, including nonnegativity.
2. Write down the KKT conditions.
3. Use complementary slackness and feasibility to identify candidate solutions.
4. Compare the objective-function values of the candidates (and any exceptional points where the linear-independence condition fails).

Existence is a separate issue

For a continuous objective on a compact, nonempty constraint set, the Extreme Value Theorem guarantees a maximizer.

Tutorial Q1: Knowledge used

Knowledge used

- **Level curves**: move the objective's level curve towards the highest feasible level (graphical approach).
- **KKT conditions**: stationarity inequalities for $x, y \geq 0$, feasibility, and complementary slackness (Lecture 11).
- **Necessity**: with affine constraints, every solution satisfies the KKT conditions.
- **Existence**: a continuous objective on a compact feasible set attains a maximum (Extreme Value Theorem).

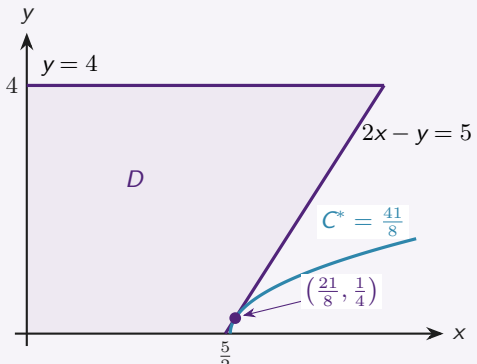
Tutorial Q1(a): Graphical approach

Solution

$$2x - 2y^2 = C \quad \Longleftrightarrow \quad x = y^2 + \frac{C}{2}.$$

Tutorial Q1(a): Graphical solution

Solution: Feasible set and optimal level curve



The highest feasible level curve touches the boundary at $(x^*, y^*) = (21/8, 1/4)$.

Tutorial Q1(b): KKT setup

Solution: Write the conditions

Let $g_1(x, y) = 2x - y - 5$ and $g_2(x, y) = y - 4$. The KKT conditions are

$$\begin{aligned} 2 - 2\lambda_1 \leq 0, \quad & x \geq 0, \quad x(2 - 2\lambda_1) = 0, \\ -4y + \lambda_1 - \lambda_2 \leq 0, \quad & y \geq 0, \quad y(-4y + \lambda_1 - \lambda_2) = 0, \\ 2x - y - 5 \leq 0, \quad & \lambda_1 \geq 0, \quad \lambda_1(2x - y - 5) = 0, \\ y - 4 \leq 0, \quad & \lambda_2 \geq 0, \quad \lambda_2(y - 4) = 0. \end{aligned}$$

Tutorial Q1(b): First deductions

Solution: Deduce $x > 0$ and $\lambda_1 = 1$

Since $2 - 2\lambda_1 \leq 0$, we have $\lambda_1 \geq 1 > 0$. Therefore, complementary slackness gives

$$2x - y - 5 = 0 \quad \Longleftrightarrow \quad 2x - y = 5.$$

If $x = 0$, this requires $y = -5$, contradicting $y \geq 0$. Hence $x > 0$ and $x(2 - 2\lambda_1) = 0$ implies

$$\lambda_1 = 1.$$

Tutorial Q1(b): Can $y = 0$?

Solution: Show that $y > 0$

Suppose $y = 0$. The second KKT derivative inequality becomes

$$-4(0) + 1 - \lambda_2 \leq 0 \quad \Longrightarrow \quad \lambda_2 \geq 1 > 0.$$

But then $\lambda_2(y - 4) = 0$ requires $y = 4$, contradicting $y = 0$.

Therefore $y > 0$.

Thus the second variable complementary slackness equation yields

$$-4y + 1 - \lambda_2 = 0.$$

Tutorial Q1(b): Cases for λ_2

Solution: Examine both possibilities

Case 1: $\lambda_2 = 0$. Then

$$-4y + 1 = 0 \;\Rightarrow\; y = \tfrac{1}{4}, \qquad 2x - y = 5 \;\Rightarrow\; x = \tfrac{21}{8}.$$

This satisfies all KKT conditions and $y < 4$. **Case 2:** $\lambda_2 > 0$. Then $y = 4$, and

$$-4(4) + 1 - \lambda_2 = 0 \quad \Longrightarrow \quad \lambda_2 = -15 < 0,$$

a contradiction.

Tutorial Q1(b): Conclusion

Solution

The only point satisfying the KKT conditions is

$$(x^*, y^*, \lambda_1^*, \lambda_2^*) = \left(\frac{21}{8}, \frac{1}{4}, 1, 0\right).$$

The feasible set is closed and bounded and the objective is continuous, so a maximizer exists. Since KKT conditions are necessary, the only KKT candidate is the **unique global solution**:

$$\boxed{(x^*, y^*) = \left(\frac{21}{8}, \frac{1}{4}\right)}.$$

Tutorial Q2

Question 2

A consumer has utility

$$u(x, y) = \sqrt{xy}, \qquad x, y \geq 0.$$

Good X costs \$5 per unit, good Y costs \$10 per unit, and income is \$500. Formulate and solve the consumer's utility-maximization problem.

Tutorial Q2: Knowledge used

Knowledge used

- **Monotonic transformation**: a strictly increasing transformation preserves maximizers (Lecture 8).
- **KKT conditions**: use derivative inequalities and complementary slackness for $x, y \geq 0$ with a budget constraint.
- **Necessity**: affine constraints guarantee that all maximizers satisfy KKT (Lecture 11).
- **Existence and comparison**: compact budget set + continuity imply an optimum exists, then compare KKT candidates.

Solution: Equivalent problem

The original problem is

$$\max_{x,y\geq 0} \sqrt{xy} \quad \text{subject to} \quad 5x + 10y \leq 500.$$

Since the square-root function is strictly increasing, this has the same solutions as

$$\boxed{\max_{x,y\geq 0} xy \quad \text{subject to} \quad 5x + 10y \leq 500.}$$

Tutorial Q2: KKT conditions

Solution: Write the conditions

$$
\begin{aligned}
y - 5\lambda \leq 0, \quad & x(y - 5\lambda) = 0, \quad & x \geq 0, \\
x - 10\lambda \leq 0, \quad & y(x - 10\lambda) = 0, \quad & y \geq 0, \\
5x + 10y - 500 \leq 0, \quad & \lambda(5x + 10y - 500) = 0, \quad & \lambda \geq 0.
\end{aligned}
$$

We next distinguish the two cases $\lambda = 0$ and $\lambda > 0$.

Tutorial Q2: The case $\lambda = 0$

Solution: Candidate at the origin

If $\lambda = 0$, the KKT derivative inequalities become

$$y \leq 0, \qquad x \leq 0.$$

Together with $x, y \geq 0$, this gives $x = y = 0$.

$$(x^*, y^*, \lambda^*) = (0, 0, 0)$$

is a KKT candidate.

Tutorial Q2: The case $\lambda > 0$

Solution: Solve the binding system

If $\lambda > 0$, then

$$5x + 10y = 500.$$

At least one good must be consumed. If $x > 0$, then $y = 5\lambda > 0$; if $y > 0$, then $x = 10\lambda > 0$. Hence both are positive.

$$y = 5\lambda, \qquad x = 10\lambda.$$

Substituting into the budget constraint gives

$$50\lambda + 50\lambda = 500 \quad \Rightarrow \quad \lambda = 5.$$

Thus $(x^*, y^*, \lambda^*) = (50, 25, 5)$ is a KKT candidate.

Tutorial Q2: Compare candidates

Solution: Unique optimum

The budget set is nonempty, closed, and bounded; v is continuous. Hence a solution exists. Compare the transformed objective:

$$v(0,0) = 0, \qquad v(50,25) = 50 \cdot 25 = 1250.$$

Therefore, the original consumer problem has the unique solution

$$\boxed{(x^*, y^*) = (50, 25)}.$$

The original utility is $u(50,25) = \sqrt{1250}$.

Tutorial Q3

Question 3

Find a solution to the maximization problem

$$\max_{(x_1,x_2)\in\mathbb{R}^2} \left(-x_1^2 - x_1x_2 - x_2^2\right)$$

subject to

$$-x_1 + 2x_2 \geq 1, \qquad 2x_1 + x_2 \leq 2.$$

Hint: Use the affine-composition result from Tutorial 8 to prove concavity.

Recap: Sufficient Lagrange conditions

Inequality-constrained maximization

Suppose f is continuously differentiable and **concave**, while every g_j is continuously differentiable and **convex**. If a feasible $\boldsymbol{x}^*$ and $\boldsymbol{\lambda}^* \geq \mathbf{0}$ satisfy

$$\boldsymbol{\nabla} f(\boldsymbol{x}^*) - \sum_{j=1}^{m} \lambda_j^* \boldsymbol{\nabla} g_j(\boldsymbol{x}^*) = \mathbf{0},$$

$$g_j(\boldsymbol{x}^*) \leq 0, \qquad \lambda_j^* g_j(\boldsymbol{x}^*) = 0 \quad (j = 1, \ldots, m),$$

then $\boldsymbol{x}^*$ is a **global maximizer**.

Recap: Sufficient KKT conditions

Nonnegative variables

Suppose f is concave and continuously differentiable, and all the g_j are convex and continuously differentiable. If a pair $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) \geq \mathbf{0}$ satisfies all KKT conditions, then $\boldsymbol{x}^*$ solves

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \quad \text{subject to } g_j(\boldsymbol{x}) \leq 0 \ \forall j, \quad \boldsymbol{x} \geq \mathbf{0}.$$

Key distinction

Necessary conditions identify candidates; **sufficient** conditions establish global optimality without a separate comparison.

Recap: Equality constraints and sufficiency

Equality-constrained problem

For

$$\max_{x \in S} f(x) \quad \text{subject to} \quad g_j(x) = 0 \ (j = 1, \ldots, m),$$

Lagrange multipliers λ_j are **unrestricted in sign**.

Sufficient conditions

If f is concave and all equality-constraint functions g_j are **affine**, then a feasible candidate satisfying

$$\nabla f(x^*) - \sum_{j=1}^{m} \lambda_j^* \nabla g_j(x^*) = \mathbf{0}$$

is globally optimal.

Recap: Affine constraints – necessary and sufficient

Corollary

If f is continuously differentiable and concave and all constraint functions are affine, then the conditions are **necessary and sufficient**:

- For equality constraints: the Lagrange conditions with unrestricted multipliers.
- For inequality constraints: the Lagrange conditions with nonnegative multipliers.
- With inequality constraints and $\boldsymbol{x} \geq \mathbf{0}$: the KKT conditions.

Consequence

Solving the corresponding Lagrange/KKT conditions identifies global solutions.

Tutorial Q3: Knowledge used

Knowledge used

- **Concavity:** an affine composition with a concave function is concave; sums of concave functions remain concave (Lecture 7 / Tutorial 8).
- **Affine constraints:** the Lagrange conditions are necessary (Lecture 10).
- **Sufficiency:** when the objective is concave and the constraints are affine, the Lagrange conditions are sufficient for a global maximum (Lecture 11).

Tutorial Q3: Establish concavity

Solution: Complete the square

Rewrite the objective:

$$\begin{aligned} f(x_1, x_2) &= -x_1^2 - x_1 x_2 - x_2^2 \\ &= -\left(x_1 + \frac{1}{2}x_2\right)^2 - \frac{3}{4}x_2^2. \end{aligned}$$

The first term is $h \circ g$, with

$$g(x_1, x_2) = x_1 + \tfrac{1}{2}x_2 \quad \text{(affine)}, \qquad h(z) = -z^2 \quad \text{(concave)}.$$

The second term is also concave. Hence f is concave. Both constraints are affine, so the Lagrange conditions are **necessary and sufficient**.

Tutorial Q3: Lagrange conditions

Solution: Set up the conditions

Write the first constraint as $g_1 = x_1 - 2x_2 + 1 \leq 0$ and the second as $g_2 = 2x_1 + x_2 - 2 \leq 0$. Then

$$\begin{aligned} -2x_1 - x_2 - \lambda_1 - 2\lambda_2 &= 0, \\ -x_1 - 2x_2 + 2\lambda_1 - \lambda_2 &= 0, \\ g_1 \leq 0, \quad \lambda_1 \geq 0, \quad \lambda_1 g_1 &= 0, \\ g_2 \leq 0, \quad \lambda_2 \geq 0, \quad \lambda_2 g_2 &= 0. \end{aligned}$$

Tutorial Q3: Case 1

Solution: $\lambda_1 = \lambda_2 = 0$

The stationarity equations reduce to

$$-2x_1 - x_2 = 0, \qquad -x_1 - 2x_2 = 0.$$

Their only solution is $x_1 = x_2 = 0$, which violates

$$g_1(0,0) = 1 > 0.$$

Thus **Case 1 is impossible**.

Solution: $\lambda_1 > 0,\ \lambda_2 = 0$

Now $g_1 = 0$, so $x_1 = 2x_2 - 1$. Substitution into stationarity yields

$$5x_2 + \lambda_1 = 2, \qquad 4x_2 - 2\lambda_1 = 1.$$

Solving,

$$x_2 = \frac{5}{14}, \qquad \lambda_1 = \frac{3}{14}, \qquad x_1 = 2 \cdot \frac{5}{14} - 1 = -\frac{2}{7}.$$

This point is feasible and satisfies all conditions:

$$(x_1, x_2, \lambda_1, \lambda_2) = \left(-\frac{2}{7}, \frac{5}{14}, \frac{3}{14}, 0\right).$$

Tutorial Q3: Case 3

Solution: $\lambda_1 = 0,\ \lambda_2 > 0$

Now $g_2 = 0$, so $x_2 = 2 - 2x_1$. Substitution gives

$$-2x_1 - (2 - 2x_1) - 2\lambda_2 = 0,$$
$$-x_1 - 2(2 - 2x_1) - \lambda_2 = 0.$$

The first equation forces $\lambda_2 = -1 < 0$, contradicting $\lambda_2 > 0$.

Case 3 is impossible.

Tutorial Q3: Case 4

Solution: $\lambda_1 > 0,\ \lambda_2 > 0$

Both constraints bind:

$$x_1 - 2x_2 = -1, \qquad 2x_1 + x_2 = 2.$$

Solving gives

$$x_1 = \frac{3}{5}, \qquad x_2 = \frac{4}{5}.$$

But then the first stationarity condition would require

$$-2\left(\frac{3}{5}\right) - \frac{4}{5} - \lambda_1 - 2\lambda_2 = 0,$$

which is impossible for $\lambda_1, \lambda_2 > 0$. Thus **Case 4 is impossible**.

Tutorial Q3: Global solution

Solution

Only Case 2 produces a point satisfying all the Lagrange conditions:

$$(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*) = \left(-\frac{2}{7}, \frac{5}{14}, \frac{3}{14}, 0\right).$$

By the sufficiency theorem, the **unique global maximizer** is

$$\boxed{\boldsymbol{x}^* = \left(-\frac{2}{7}, \frac{5}{14}\right)}.$$

Key Takeaways

- Use KKT conditions when decision variables must be nonnegative.
- Begin with complementary slackness to identify binding constraints and feasible cases.
- Necessity screens candidates; existence plus comparison identifies solutions.
- With a concave objective and affine constraints, Lagrange/KKT conditions are also sufficient for global optimality.

Assignment 3: Question 1

Question 1. Miss P can bake 2 kinds of cakes, cake A and cake B. Cake A requires 450 grams of egg, 400 grams of flour and 200 grams of milk. Cake B requires 450 grams of egg, 1000 grams of flour and 100 grams of milk. Miss P only has 4.5 kilograms of egg, 7 kilograms of flour and 1.6 kilograms of milk. Miss P is able to make \$60 of profit for cake A and \$120 of profit for cake B.

(a) Draw the constraint set. (3 marks)

(b) Present the standard form of the maximisation problem. (2.5 marks)

(c) Does this problem have a solution? Justify why or why not. (2.5 marks)

(d) How many of each type of cakes should Miss P bake to maximise her profit? Also what is her profit? (7 marks)

(e) At the optimal solution in part (d), which constraints are binding and which are slack? (3 marks)

(f) What is the corresponding dual program? (4 marks)

(g) Interpret the dual variables using the context of this question. (3 marks)

Assignment 3: Question 1– Hints

Hints

- **(a) Constraint set:** *Refer to Tutorial 9, Q2(b).*
- **(b) Standard form:** *Refer to Tutorial 9, Q2 (formulation).*
- **(c) Existence:** EVT *Refer to Tutorial 8, Q3; Tutorial 9, Q2(a).*
- **(d) Optimal production:** Fundamental Theorem of Linear Programming. *Refer to Tutorial 9, Q2(b) and Q3(a).*
- **(e) Binding/slack.**
- **(f) Dual.** *Refer to Tutorial 9, Q2(c) and Q3(b).*
- **(g) Interpretation.** *Refer to Tutorial 9, Q2(d) and Q3(c).*

Assignment 3: Question 2

Question 2. In finance, various risk and return metrics can be aggregated to define a total portfolio performance metric $F(\boldsymbol{x})$. Suppose we have k distinct component metrics, $f_1, f_2, \ldots, f_k : S \to \mathbb{R}$, where each S is the convex set of feasible asset allocations. We define the total portfolio performance metric as the sum function:

$$F(\boldsymbol{x}) = \sum_{i=1}^{k} f_i(\boldsymbol{x}).$$

Prove that if every component metric f_i is a convex function on S, then the total portfolio performance metric F is also a convex function. Your proof must rely solely on the definition of convexity and the properties of summation, without assuming differentiability or using second-order conditions. (7 marks)

Assignment 3: Question 2– Hints

Hints

- Start with arbitrary $\boldsymbol{x}, \boldsymbol{y} \in S$ and $\lambda \in [0, 1]$. Write the convexity inequality for *each* component f_i.
- Use the properties of summation to combine those inequalities, then compare the resulting statement with the definition of convexity for F. *Refer to Tutorial 7, Q2.*

Assignment 3: Question 3

Question 3. Use the Extreme Value Theorem to show that the following problems have a solution:

(a)

$$\min_{(x,y)\in D} f(x,y) = \sqrt{x^2+y^2}, \qquad D = \{(x,y) \in \mathbb{R}^2 : |x| + |y| \leq 2\}.$$

(7 marks)

(b) Draw the associated constraint set D from part (a). (3 marks)

Note: Be sure to argue rigorously and appeal to the presentation from real analysis (first part of the course).